1.5b

Sleep
Sleep stages & theories

What is sleep?

Humans wake for ~16 hours before sleeping.

Sleep is a periodic loss of conciousness, distinct from comas, hibernation, anesthesia, etc.

By recording brain waves & muscle movements of sleeping people, observing & occasionally waking them, researchers can study sleep.

How does our biology follow rhythyms?

Our bodies ~synchronize w/ 24hr day using our internal biological clock :: circadian rhythm

morning nears ⇒ body temp ↑. It peaks during the day, dips in early afternoon, drops during evening.

Thinking is sharpest & memory most accurate during daily peak in circadian arousal.

Personal connection

Because of my circadian rhythm, during many all nighters, I tend to feel groggy during the night & alert when its time for school.

Age & experience can change our CRs.

20yr olds are evening-energized, performance ↑ throughout the day; "owls"

Older adults experience more fragile sleep, and are morning-loving "larks". For our hunter-gatherer ancestors, a grandparent waking early helped protect family from predators.

After age 20 (slightly earlier in women) we gradually shift from owls to larks.

Night owls tend to be smarter & more creative

Morning types tend to do better in school, take more initiative, less vulnerable to deppression

What is the biological rhythm for sleep?

Every ~90 min, you cycle through sleep stages

Periods of fast jerky eye movement also had energetic brain activity ∴ REM sleep

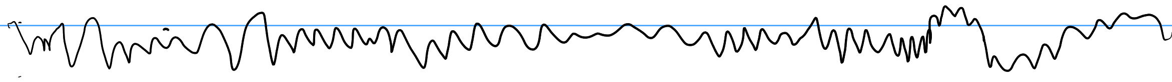
To measure, when ur ready for sleep, researchers strap electrodes to:
scalp - to detect brain waves
chin - to tetect muscle tension
just outside eye corners - to detect eye movements

Other devices may record Heart rate, respiration rate & genital arousal.

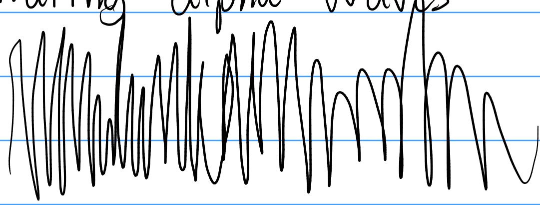
While you are in bed, awake with closed eyes, EEG shows relatively slow alpha waves

Eventually, you slowly drift to sleep. This transition is marked with slowed breathing and irregular brain waves of N1 sleep, a part of NREM sleep.

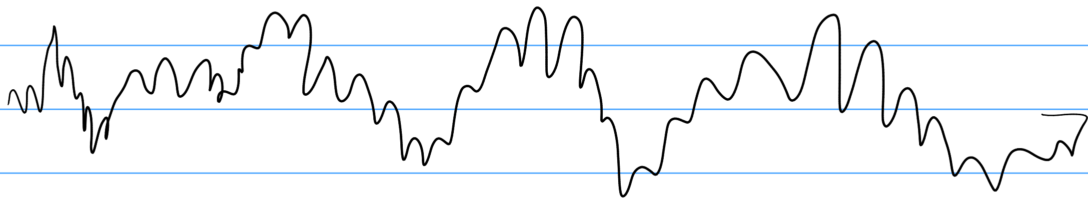
Waking Beta waves



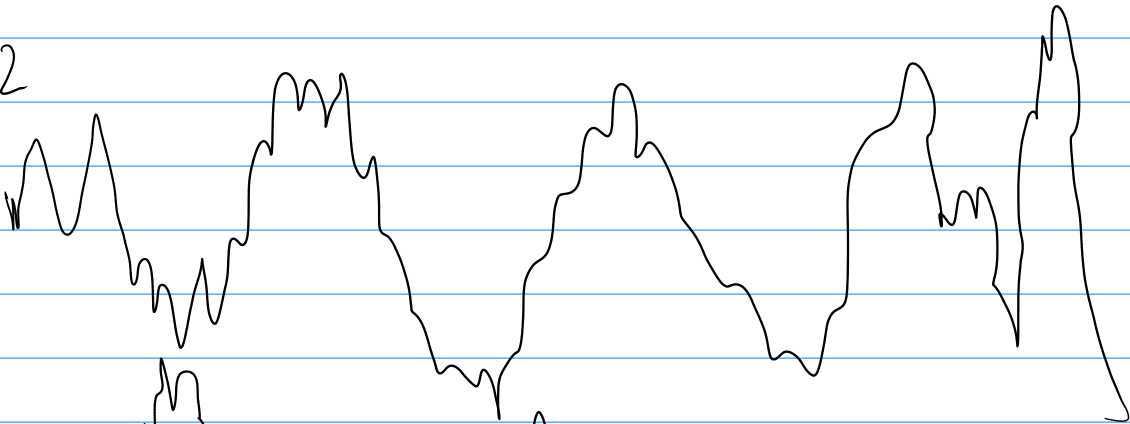
Waking alpha waves



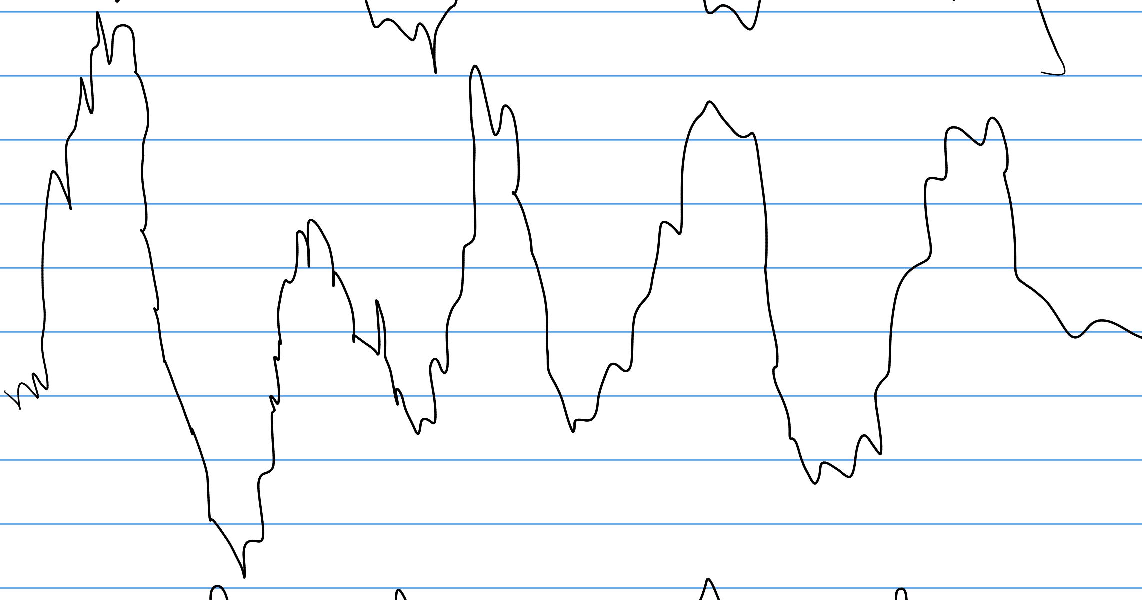
N1.



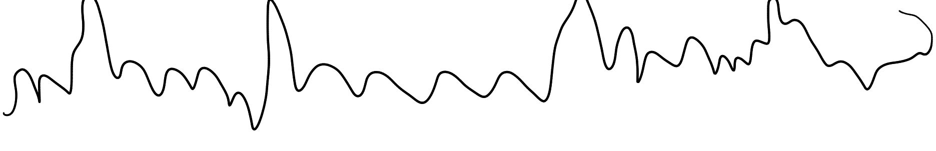
N2



N3



REM



Sleep deprived man asked to press button every time strobe lights flashed in his taped open eyes. after a few min, he missed one bc he fell asleep for 2 sec, without realizing.

During brief N1 sleep you may experience hallucinations :: sensory experience without stimuli
You may have sensation of falling or of floating weightlessly

These hypnagogic sensations may be incorporated into memories

After stage N1, you begin ~ 20 min of stage N2 sleep -

periodic sleep spindles :: bursts of rapid, rhythmic brain - wave activity that aid memory processing

you can be easily awakened, but are clearly asleep

Then you transition to the deep sleep of stage 3, for ~30 min

Brain emits large, slow, delta waves

you are hard to awaken

end of this stage, children may bedwet

~1 hour after falling asleep, you return through N2 (where you'll spend ~½ of night) entering REM.

for ~10 min, brain waves become rapid & saw-toothed.

Heart rate ↑, breathing becomes rapid

~every ½ min, eyes momentarily dart in bursts of activity, behind closed lids.

These eye movements announce beginning of a dream.

often emotional, usually story-like, and richly hallucinatory

Unless very scary, genitals get aroused during REM

In REM, motor cortex is active, but brain stem blocks messages.

So muscles relaxed so much, other than occasional twitch, or essentially paralyzed.

(This immobility may linger, → sleep paralysis)

REM is aka paradoxical sleep.

Body is internally aroused, yet externally calm.

The cycle repeats ~every 90 min for younger adults, shorter, more frequent cycles for older adults.

As night goes on, deep N3 grows shorter & disappears, REM & N2 get longer.

By morning 20-25% of avg sleep, ~100 min, in REM

In studies, 37% report rarely or never remembering dreams.

But, ≥80% of the time, even they could remember dreams when woken during REM.

How does our biology & environment affect sleep

Newborns sleep for 2/3 of day

most adults sleep ≤ 1/3 of day, but can vary

One analysis of 1.3m people identified 956 genes related to sleep patterns like insomnia. another associated genes with being a morning person.

Sleep patterns are also culturally, & economically influenced.

In Canada, Britain, Germany, Japan, US, adults average 7 hours sleep on work days & 7-8 hours on other days

Earlier school start times, more extracurriculars, fewer parent set bedtimes lead american kids to get less sleep than australians.

Stress, discrimination & Poverty are also Major Factors for sleep

Bright light affects sleepiness by activating light sensitive retinal proteins.

This signals the brain's suprachiasmatic nucleus to decrease production of melatonin: sleep inducing hormone found in hypothalamus

Covid pandemic's lockdown led many to experience lower than normal light levels.
Night shift workers may experience chronic desynchronization. ∴ More likely to develop fatigue, stomach problems, heart disease, breast cancer.

90% of americans report using light-emitting device 1 hr before bed.
This artificial light delays sleep & affects sleep quality

Streaming affects dreaming

It can be hard to try & sleep in on weekends & stay up during the day.

People doing this experience a kind of jet lag

Why do we sleep?

Sleep protects.

Our ancestors were better off in a cave at night instead of outside in harms way.

A species sleep pattern tends to suit its ecological niche.

Sleep restores

Sleep gives body & brain a chance to repair, rewire & reorganize.

Helps body heal from infection, restores immune system

Gives resting neurons time to repair while pruning unused connections

Sleep sweeps away protein fragments that can cause Alzheimers.

Sleep aids memory consolidation

Restore & rebuild our fading memory of day's experiences

Memories are consolidated during slow wave deep sleep by replaying recent learning & strengthening neural connections

Reactivates recent experiences stored in hippocampus, and moves them to permanent storage.

Adults and children trained in a task perform it better after a night's sleep, or a short nap, rather than a few hours awake.

Sleep feeds creative thinking

Dreams can inspire noteworthy artistic & scientific achievements.

Sleep gives a boost to our thinking & learning.

Sleep supports growth

During slow wave sleep, pituitary gland releases human growth hormone - nessesary for muscle development

Sleep conserves energy

Preserves our energy for waking times, when gathering, foraging & hunting would have happened.

MCQ: 1. b 2. c 3. b 4. b. 5. b 6. d